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Computer Networking

Computer Networking is the practice of connecting two or more computers or devices so they can share information, resources, and services with each other.

- Networking is the feature of an Operating System
- Any Operating System which is built based on OSI model can provide networking facilities.

Terms used in Computer Networking

Protocols
A set of rules to communicate on a network (HTTP, DHTP).

Node
A Device on the network is called Node

Transmission
Sending data signals over a medium (cable, WIFI)

Data Communication
covers the entire process of exchange data, including transmission, reception, and interpretation of data.

Birth of internet

In 1969, ARPANET, the first computer network, was created by the U.S. Department of Defense. Its primary purpose was to ensure communication during a nuclear war. ARPANET initially connected four universities, but by 1972, it expanded to include 40 machines. ARPANET laid the foundation for the modern Internet.

Types of Transmission

1. **Simplex**

Simplex Communication is one-way communication where one device sends data, and the other only receives.

2. **Duplex**

Duplex Communication is Two-way communication where both devices can send and receive Data.

- Half Duplex

  Only one device can send data at a time.

  Example: Walky-talky, Radio

- Full Duplex

  Both devices can send and receive data at the same time.

  Example: Voice Call, Video Call

Synchronous Transmission

When Sender and receiver both must me online to communicate.

Example: Voice Call

Asynchronous Transmission

When the receiver doesn't necessarily have to be online for communication.

Example: SMS

Transmission media

The physical pathway used to transmit data is called transmission media. And its classified into two categories.

GUIDED MEDIA

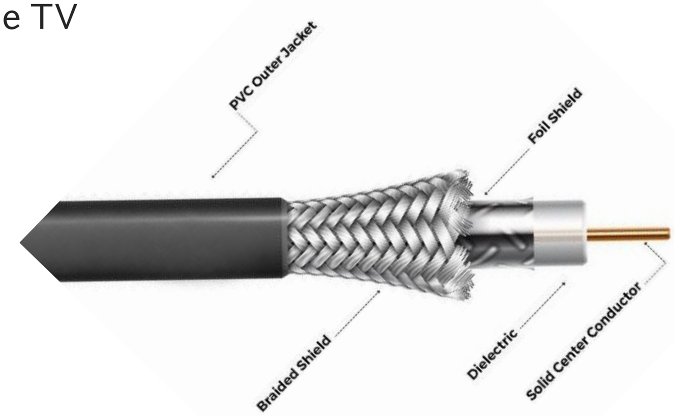
The media which transmit data through a fixed path such as cables and wires.

Coaxial Cable

A cable consists of a central conductor, an insulating layer, a metal shield, and an outer cover. It transmits data using electrical signals. Commonly used for cable TV

Connectors used:

- BNC Connector
- T Connector
- END Connector



Fiber Optic Cable Made of glass or plastic fibers; this cable transmits data in the form of light signa ls. Fiber optic cables are known for their high speed and capacity

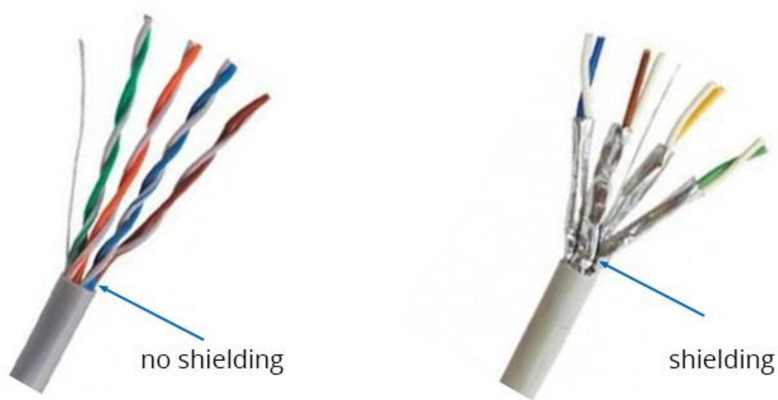
Twisted Pair Cable

This cable consists of pairs of insulated wires twisted together.

- **Shielded Twisted Pair (STP)** These cables have a braided shield around them, offering protection against external interference.

- **Unshielded Twisted Pair (UTP)** These cables lack the shielding.

- **Categories:** Cat5, Cat6, Cat7 etc. with higher categories offering better performance and faster data transmission.



UNGUIDED MEDIA

The Media which transmits data through free space**.**

Microwave

- Definition: Microwaves are high-frequency radio waves.
- Range: 1 GHz to 300 GHz.
- Use: Satellite communication, Wi-Fi, mobile networks, radar.
- Feature: Can carry large amounts of data over long distances but need line-of-sight.

Radio Wave

- Definition: Radio waves are the lowest-frequency electromagnetic waves.
- Range: 3 Hz to 300 GHz (very broad).
- Use: AM/FM radio, TV broadcasting, mobile phones, walkie-talkies.
- Feature: Can travel long distances and penetrate buildings.

Infrared

- Definition: Infrared waves are just below visible light in frequency.
- Range: 300 GHz to 400 THz.
- Use: Remote controls, short-distance communication, night-vision cameras.
- Feature: Works only in short range, usually line-of-sight, can’t pass through walls.

Ground Link and Satellite Link

Ground Link

Communication that happens using cables or optical fiber on the ground. Example: Telephone lines, Ethernet cables, or fiber optic internet.

Satellite Link

Communication between Earth stations and satellites using radio or microwave signals. It covers a very large area, good for remote locations, but can have delay (latency) and be affected by weather.

Examples: TV broadcasting, GPS, Internet.

Signal Type

1. **Analog Signal**

Analog signals are continuous waves to represent information.

Example: Radio waves. Infrared, Microwave

2. **Digital Signal**

Digital signals are discrete pulses representing information in binary form (0s and 1s).

Example: Data transferred over Ethernet cables.

Protocols

HTTP (Hypertext Transfer Protocol): A protocol used for transferring web pages and resources on the internet. It works over TCP and is not secure.

HTTPS (Hypertext Transfer Protocol Secure): A secure version of HTTP that uses SSL/TLS encryption to protect data transfer between client and server.

DHCP (Dynamic Host Configuration Protocol): A protocol that automatically assigns IP addresses and other network configuration details to devices on a network.

ICMP (Internet Control Message Protocol): A protocol used by network devices to send error messages and operational information (e.g., "ping" uses ICMP).

TCP (Transmission Control Protocol): A connection-oriented protocol that ensures reliable data transfer by breaking data into packets and reassembling them in order. And both the source and destination node must be active and connected

TCP/IP (Transmission Control Protocol/Internet Protocol): A suite of communication protocols that define how devices communicate on the internet. TCP ensures reliability, while IP handles addressing and routing.

PUSH Protocol: A communication model where the server sends data to the client without the client requesting it (e.g., notifications).

PULL Protocol: A communication model where the client requests data from the server (e.g., when loading a webpage).

ARP (Address Resolution Protocol): Converts an IP address into its corresponding MAC address.

RARP (Reverse Address Resolution Protocol): Converts a MAC address into its corresponding IP address.

CSMA (Carrier Sense Multiple Access): A network protocol where devices listen to the communication channel before sending data. If the channel is busy, the device waits; if it's free, it transmits.

CSMA/CD (Carrier Sense Multiple Access with Collision Detection): Used in wired networks (like Ethernet). Devices listen before sending data, and if two devices send data at the same time (collision), they stop, wait for a random time, and retransmit.

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance): Used in wireless networks (like Wi-Fi). Devices try to avoid collisions by signaling intent to transmit before sending data, since detecting collisions in wireless is difficult.

RIP (Routing Information Protocol): A distance-vector routing protocol. Uses hop count as the metric to determine the best path.

OSPF (Open Shortest Path First): A link-state routing protocol. Uses Dijkstra’s algorithm to calculate the shortest path.

UDP (User Datagram Protocol) is a connectionless Transport Layer protocol that prioritizes speed over guaranteed delivery and error correction.

BGP (Border Gateway Protocol) is an inter-domain routing protocol used to exchange routing information between different autonomous systems on the Internet.
It selects the best path based on policies, path attributes, and network rules rather than just distance.

Network Connectivity Device

HUB

A hub is a simple networking device that connects multiple devices in a network and forwards data to all of them without filtering.

- Provides a central connection point in a LAN.
- Broadcasts incoming data to every connected device.
- All devices see the data, but only the intended device uses it.
- Works only on Logical Addresses
- Works on half Duplex
- Works only with electrical signals, not with addresses.
- Works on Layer 1 – Physical Layer
- Collision can occur

SWITCH

A switch is a network connectivity device that connects multiple devices in a LAN and forwards data only to the intended device using MAC addresses.

- Provides a central connection point in a LAN.
- Maintains a table of Physical IPs associated with Corresponding with Logical Ips of devices on the network.
- ARP protocol reads this table to find Physical IP From Logical IP
- RARP protocol reads this table to find Logical IP From Physical IP
- Works on Layer 2 – Data Link Layer
- No Collision
- Only one time broadcast to check which devices connected.

ROUTER

A router is a network connectivity Device that is used to connects multiple networks to communicate.

- Connects multiple networks (e.g., home LAN to the internet).
- Selects the best path for transferring data packets.
- Examine the IP address of incoming packets.
- Uses routing tables and protocols (like RIP, OSPF, BGP) to decide where to send the packet.
- Can perform Modulization and Demodulization
- Works on Layer 3 – Network Layer

REPEATER

A repeater is a device that regenerates and amplifies network signals to extend the transmission distance.

- Extends the physical range of wired or wireless signals.
- Prevents signal loss over long distances.
- Works on Layer 1 – Physical Layer

ACCESS POINT

An access point is a device that allows wireless devices to connect to a wired LAN using Wi-Fi.

- Provides wireless connectivity in a network.
- Extends the coverage of a wired LAN to wireless devices.

How it works:

- Connect to a switch or router via cable.
- Converts wired signals into wireless signals.
- Wireless devices connect using Wi-Fi standards.
- Works on Layer 2 – Data Link Layer

BRIDGE

A bridge is a device that connects two LAN segments and forwards traffic based on MAC addresses.

- Reads the MAC address of frames.
- Forwards or blocks traffic depending on the destination.
- Reduces collisions by segmenting the network.
- Works on Layer 2 – Data Link Layer

GATEWAY

A gateway is a device that connects two networks that use different protocols and translates communication between them.

Allows communication between networks that otherwise couldn’t communicate.

Example: connects a private LAN to the public internet.

How it works:

- Acts as an entry/exit point for a network.
- Convert data between protocols (e.g., TCP/IP ↔ VoIP, IPv4 ↔ IPv6).
- Often implemented in routers or firewalls.
- Works on Layer 3 Network Layer.

Classification of Network

CLIENT-SERVER NETWORK

A network where clients (users) are connected to a server and request services from that server and the server provides those services.

- Distributed Client-Server Network

  Where each user on the network has their own resources. Resources such as Memory, Storage, Processing power.

- Centralized Client-Server Network

  Where all the users are dumb terminals connected to a main server and the resources are provided by the main server

PEER-TO-PEER NETWORK (P2P)

A network where all devices are equal and can act as both clients and servers.

Example: Block Chain

Types of Networks According to Area

PAN (Personal Area Network)

A small network connecting personal devices within a short range.

Example: Bluetooth connections between a smartphone and wireless headphones

LAN (Local Area Network)

A network that connects devices within a limited area such as a home or office.

Example: The network in an office connecting computers and printers.

MAN (Metropolitan Area Network)

A network covering a city or a large campus.

Example: A university's network across a city.

WAN (Wide Area Network)

A network covering a large geographic area, such as a country or continent.

Example: Broadband, Zong.

GAN (Global Area Network)

A network covering the entire globe/world.

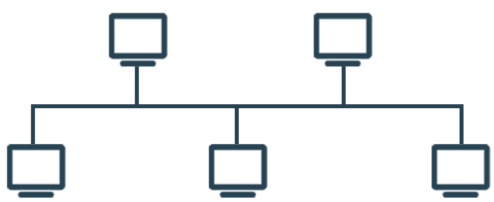
Example: The Internet

Topology

Physical arrangement of the devices (router, switch, node) on a network is called topology.

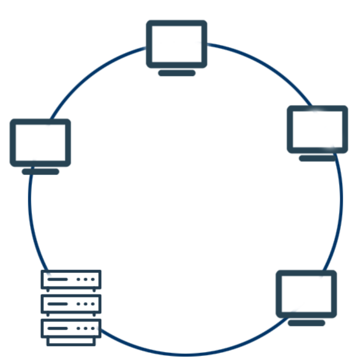
Bus Topology

All devices are connected to a single main cable (the bus). Data travels along this cable, and only one device can send at a time. If the main cable breaks, the whole network stops working.



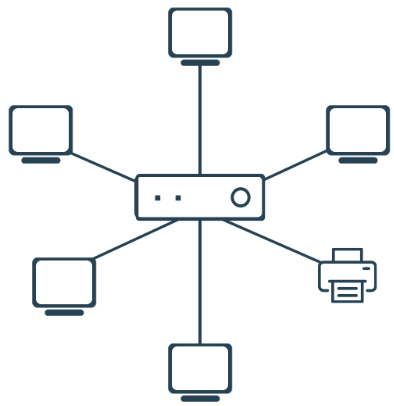
Ring Topology

Devices are connected in a circular loop. Data moves around the ring in one direction until it reaches the right device. If one device or connection fails, the whole ring may stop working.



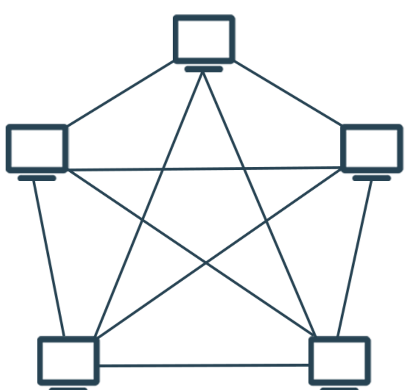
Star Topology

All devices connect to a central device (like a hub or switch). Data passes through this central device to reach others. If one device fails, the rest still works, but if the central device fails, the entire network goes down.



Mesh Topology

Every device connects directly to many or all other devices. This gives multiple paths for data, making the network very reliable. Even if one link fails, data can still travel through another path.



Addresses in Networking

MAC Address (Media Access Control Address)

This is the 'Physical Address' which is a unique hardware address assigned to a device.

- Written in hexadecimal format 00:1A:2B:3C:4D:5E
- network interface card (NIC)

IP Address (Internet Protocol Address)

This is a 'Logical Address' used to identify devices in a network, including computers, routers, and other devices.

Types of IP Address

Public IP Address

A unique IP address is assigned to devices that are directly accessible from the internet. These are typically assigned by the Internet Service Provider (ISP).

Private IP Address

An IP address used within a private network and not routable on the public internet. These are reserved for local network usage.

Versions of IP Address

IPV4 (Internet Protocol Version 4)

An IPv4 address consists of 32 bits, divided into 4 octets (each ranging from 0 to 255).

- Support a total of 2^{32} about 4 billion unique addresses.
- Example: `192.168.1.1`

IPV6 (internet Protocol Version 6)

An IPv6 addresses are 128 bits long and are written as eight groups of four hexadecimal digits, separated by colons.

- Support 2^{128} about 3 trillion unique addresses
- Example: `2001:0db8:85a3:0000:0000:8a2e:0370:7334`

Class of IPV4

Class A | N | | H | | H | | H | Range 1 – 126

Class B | N | | N | | H | | H | Range 128 – 191

Class C | N | | N | | N | | H | Range 192 – 223

Class D | Range 224 - 239 (Reserved for multicast addressing)

Class E | Range 240 - 255 (Reserved for research purposes and future use)

127 Dedicated for live server or to create a local server and loop back.

Subnetting

It is the process of dividing a single IP address into smaller subnets so multiple devices can use a single IP address

Classless Inter-Domain Routing

It is a method of writing IP addresses along with their prefix length, written as:

IP Address / Prefix Length Example: 100.0.0.1/8

The number after the slash shows how many bits of the IP address are reserved for the Network ID.

NOTE: Whenever you see any IP with prefix length less than 8, that IP is divided through subnetting.

Network Address The IP address which is taken by Switch itself.

- The IP with all host octets ‘0’ is always network address.
- Example: 192.168.1.0

Broadcast Address The address used by switch to broadcast any transmission on the network

- Take the network address and change all host octets to ‘255’
- Example: 192.168.1.255

Subnet Mask The IP where all the network octets are ‘255’ and host octets ‘0’

- Example: 255.255.255.0

ROUTING

Routing refers to the process of selecting paths in a network to send data from a source to a destination.

Dynamic Routing

Routing In dynamic routing, routes are automatically adjusted based on network changes using routing protocols like OSPF (Open Shortest Path First) and BGP (Border Gateway Protocol). These protocols help routers determine the best paths dynamically.

- **Distance Vector Routing**
  It Finds the path with the least number of routers in between.
  Algorithm used: bell ford
  Protocol used: RIP

- **Link State Routing**
  It finds the shortest path according to the weight of edges regardless of the number of routers in between.
  Algorithm used: Dijkstra
  Protocol used: OSPF

Static Routing

In static routing, routes are manually configured by the network administrator. These routes remain fixed and do not change unless updated manually.

DTE and DCE

DTE (Data Terminal Equipment)

These are devices that serve as endpoints in data communication networks, such as computers, printers, or other devices that generate or consume data. These devices communicate through data interfaces.

DTE Port is the port which doesn’t have any clock rate and is used for receiving.

DCE (Data Communication Equipment)

These are devices that facilitate data transmission between DTEs. Examples include modems, routers, and switches. DCE devices convert data formats and manage communication between DTEs.

DCE Port is the port which has a clock rate and is used for sending.

OSI MODAL

OSI Modal / (Open System Interconnect) Modal is a theoretically oriented model. It is ISO (Internation Standardization Organization) certified, and to provide networking capabilities for an operating system, it needs to work with OSI Modal.

NOTE: OSI Modal is a theoretical oriented modal, where the TCP / IP Modal is the practical oriented model.

OSI LAYERS

1. PHISICAL LAYER

This layer is responsible for the physical connections and between nodes, which Includes cables, connectors, etc.

- The HUB works on the Physical Layer.
- Data is converted in bits on this layer.

2. DATA LINK LAYER

On this layer, Transmission within the network is performed.

- Here Data is assigned Receiver's Mac Address which then is called Frame.
- Here Error Detection, Error Correction and Flow Control is performed.
- Here, Point-to-Point checking of transmission is done by using the CRC (Cyclic Redundancy Check) algorithm.
- The Switch works on this layer along with ARP and RARP protocols.

3. NATWORK LAYER

On this layer, transmission from network to networks is performed.

- Here the Frames are assigned the receiver's IP which then are called Packets.
- The path for transmitting the packet is decided on this layer.
- The Router works on this layer along with Protocols such as RIP and OSPF.

4. TRANSPORT LAYER

This layer is a segment between hardware layers and software layers.

- This layer breaks the data into segments called chunks and assembles it on the receiver's node.
- It ensures that the communication is end-to-end.
- It ensures that the data received is complete, and if it isn't, then it generates a request to retransmit it. This process is done using TCP protocol.

5. SESSION LAYER

This layer is to create a session between two nodes, enabling communication and transmission.

- The session layer also synchronizes data transfer with checkpoints so if error occurs, the transmission will continue from last checkpoint, not from the beginning.
- The Sender's Session layer communicates with the Receiver's Session layer to create the session.

6. PRESENTATION LAYER

This layer prepares and translates the data to make the data representable for application layer.

- It encrypts data on the sender application and decrypts it on the receiver's application.
- It compresses the data to reduce the overall size of the data transferred.

7. APPLICATION LAYER

This layer represents the data to the user on the application, software like web browser and email rely on this layer to establish communication on the network.

NOTE: The web browser software or the email software are not part of the application layer but it is the protocol on which software relies on to establish communication on the network. Some of its examples are HTTP and SMTP.

TCP/IP MODEL

TCP/IP reduced the 7 layers of OSI Model to four layers.

APLICATION LAYER	• Application • Presentation • Session
TRANSPORT LAYER	
NETWORK LAYER	
INTERNAL LAYER	• Datalink • Physical

MULTIPLEXING

Multiplexing is a technique used in networking where multiple signals/data streams are combined and sent over a single communication channel.

TYPES OF MULTIPLEXING

1. ANALOG MULTIPLEXING

Used for analog signals (old telephone lines, radio signals).

- **Frequency Division Multiplexing (FDM)**

  The channel is divided into different frequency bands, and each user gets a unique frequency range. All signals are sent at the same time, but at different frequencies.

  Used in Radio broadcasting, Cable TV, Traditional telephone lines etc.

- **Wavelength Division Multiplexing (WDM)**

  Special type of FDM for fiber optics, uses different light wavelengths (colors) to send multiple data streams through one optical fiber.

  Used in Fiber optic communication and High-speed internet backbone.

2. **DIGITAL MULTIPLEXING**

Used for digital signals (networking, digital communication).

- **Time Division Multiplexing (TDM)**
  The channel is divided into time slots, and each user gets their own time slot to send data.
  - **Synchronous TDM**

    Each user gets fixed time slots, even if they have no data to send → sometimes wastes time.
  - **Asynchronous TDM**
    Time slots are given dynamically only to users who have data. No time is wasted.

- **Code Division Multiplexing (CDM) / CDMA**

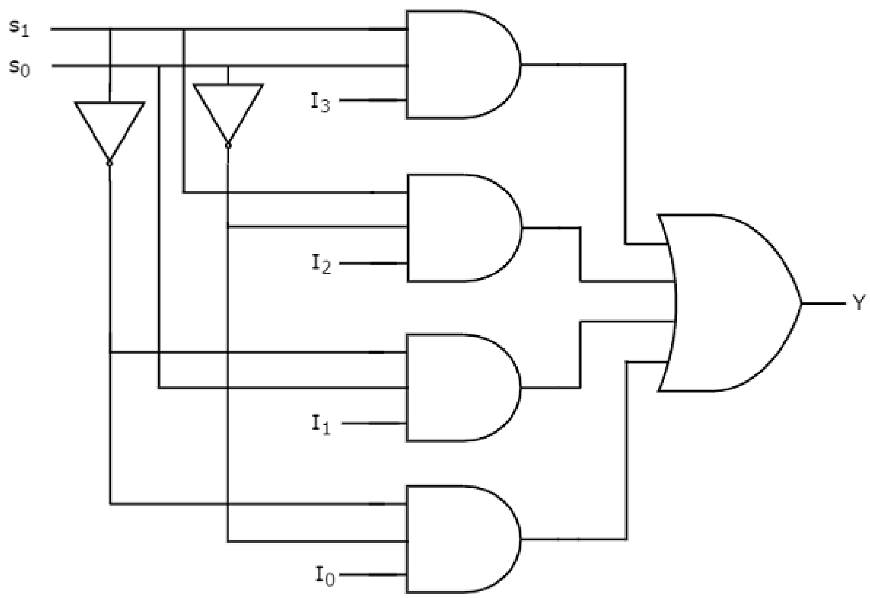
  All users send data at the same time, but each user gets a unique code, and Receiver uses code to extract the correct data

  Used in 3G mobile networks and GPS

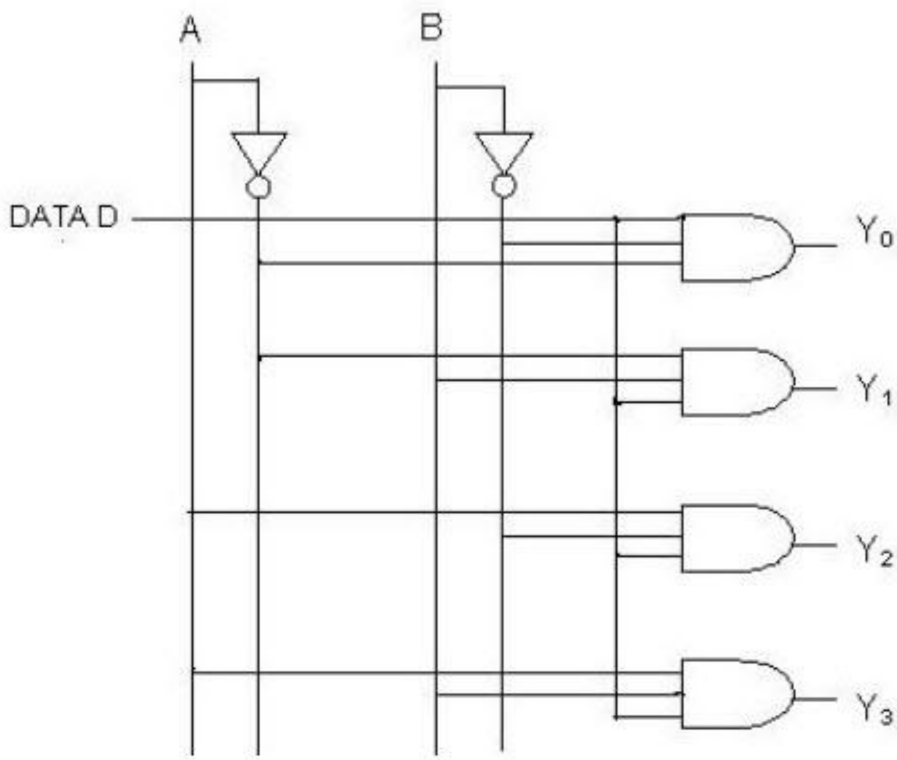
DEMULTIPLEXING

Demultiplexing is the process of taking a single, combined input signal and separating it back into the multiple individual, original signals that were originally bundled together.

MULTIPLEXER CIRCUIT



DEMULTIPLEXER CIRCUIT

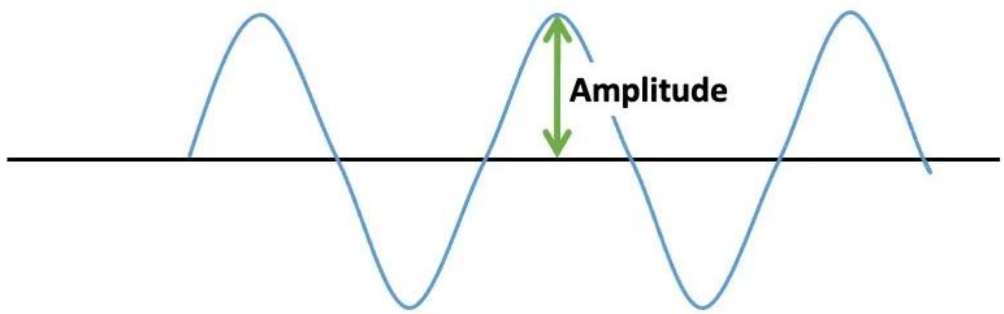


MODULATION

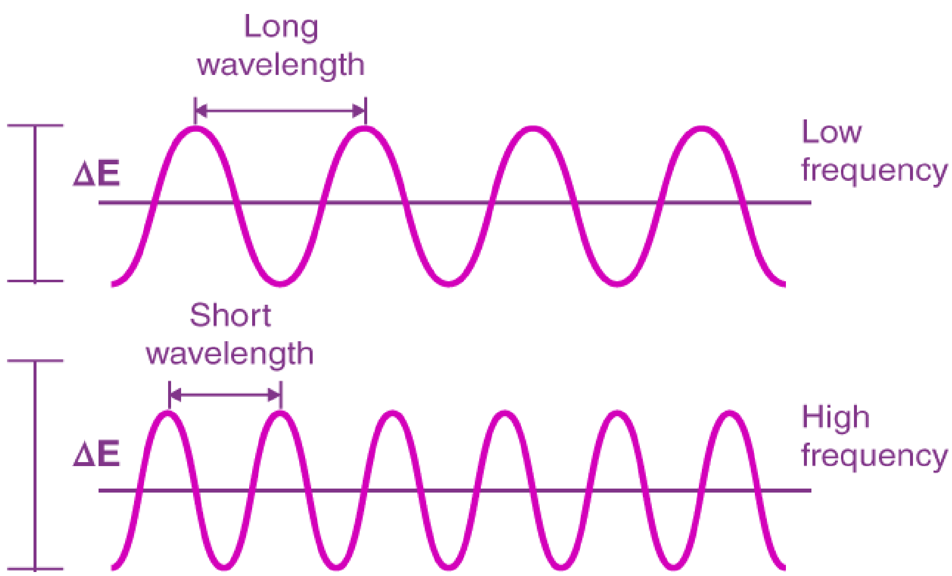
Modulation is the process of adding digital or analog information to a carrier signal so it can be transmitted over long distances.

Modulation means changing a carrier signal's properties (amplitude, frequency, or phase) to send data.

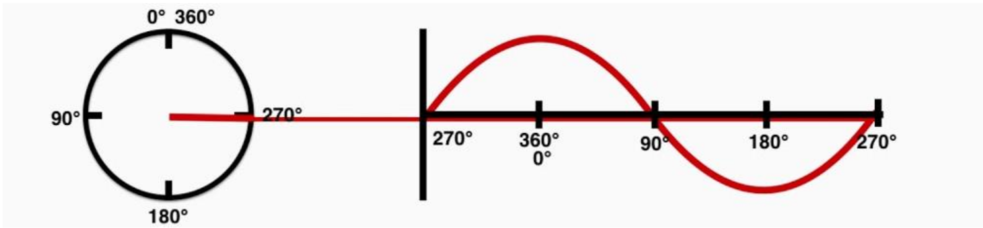
Amplitude: The height of the wave from its center position. It dictates the strength of the signal.



Frequency: The number of complete wave cycles that occur per second. It determines the pitch or channel of the signal.



Phase: The initial angle or starting point of the wave at a specific time. It represents the position of the wave.



A Modem is a device that performs both modulation and demodulation processes.

Carrier Signal: The signals that contain no information but have a certain phase, frequency, and amplitude are called carrier signals.

Message Signal: The signals that contain the information to be transmitted; they are used to modulate the carrier signal's properties. It is also called (baseband signal or Modulating Signal)

Modulated Signal: The signals which are obtained after modulating the carrier signals according to Massage signals are called Modulated Signals.

Advantages of Modulation

- It reduces the size of the antenna.
- It reduces the cost of wires.
- It prohibits the mixing of signals.
- It increases the range of communication.
- It improves the reception quality.
- It easily multiplexes the signals.

TYPES OF MODULATION

1. ANALOG MODULATION

Used when message signal is analog.

- **Amplitude Modulation (AM)**
  Amplitude of carrier wave is changed, and Frequency & phase remain constant.
  Used for: AM radio, Aircraft communication systems

- **Frequency Modulation (FM)**

  Frequency of carrier wave is changed, and Amplitude & phase remain constant

  Used for: FM radio, TV audio transmission

- **Phase Modulation (PM)**
  Phase of carrier wave is changed, and Amplitude & frequency remain constant
  Used for: Satellite communication

The purpose of changing the Amplitude, Frequency, and Phase of the Carrier signal is to add the data of Massage signal into it.

2. DIGITAL MODULATION

Used when message signal is digital.

- **ASK (Amplitude Shift Keying)**
  The Amplitude of Carrier signal is modulated based on bit (0 or 1) of Massage Signal

  Digital 1 = high amplitude

  Digital 0 = low amplitude

- **FSK (Frequency Shift Keying)**
  The Frequency of Carrier signal is modulated based on bit (0 or 1) of Massage signal

  Digital 1 = high frequency

  Digital 0 = low frequency

- **PSK (Phase Shift Keying)**

  The Phase of Carrier Signal is Modulated based on bit (0 or 1) of Massage Signal

  - **BPSK** – 2 phase shifts
  - **QPSK** – 4 phase shifts
  - **8-PSK** – 8 phase shifts

- **QAM (Quadrature Amplitude Modulation)**

  It is the combines **ASK** and **PSK**. Purpose of it is to modulate the Amplitude and Phase Simultaneously, allowing it to pack more digital information into a single symbol. Used in Wi-Fi and 5G Networks.

DEMODULATION

Demodulation is the process of extracting the original information from a Modulated signal. The modem will convert the carrier variation of amplitude, frequency, and phase back to the original message signal at the receiver's end.